

Academic Curriculum & Syllabus Book

ELECTRONICS & COMMUNICATION ENGINEERING

for

B. TECH. FOUR YEAR DEGREE PROGRAM

(Applicable for the batches admitted from A.Y. 2024-25)



ADITYA

UNIVERSITY

Department of Electronics & Communication Engineering
B.Tech Program Curriculum-2024



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Department of Electronics & Communication Engineering

B.Tech (ECE) Program Curriculum-2024

(Applicable for the batches admitted from the A.Y. 2024-25)

UG Programs Offered

- B. Tech in (Electronics & Communication Engineering)
- B. Tech in (Electronics & Communication Engineering) with
 - Minor degree in Civil Engineering
 - Minor degree in Electrical and Electronics Engineering
 - Minor degree in Mechanical Engineering

Minor Streams offered in B.Tech (Electronics & Communication Engineering)

- Minor streams will be listed after configuration.

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S.No	Broad Category of Course	UGC	Credits
1	Ability Enhancement Courses (AEC)	8	0
2	Mandatory Courses (MC)	-	0
3	Major Core Courses (MCC)	80	16
4	Multidisciplinary Courses (MDC)	9	0
5	Minor Stream Courses (MSC) (or) University Open Elective Courses (UEC)	32	0
6	Full Semester Internship (PROJ)	12	0
7	Skill Enhancement Courses (SEC)	9	4
8	Summer Internships (SI)	2-4	0
9	Value Added Courses (VAC)	6-8	0
Total Credits to be earned for B. Tech Degree		160	24

Foundation Courses - FC

Intermediate-level Courses - IC

Advanced Courses - AC

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Applied Chemistry	MCC	3		1	4
Data Structures	MCC	3	1		4
Object Oriented Programming	MCC	3		1	4
Database Management Systems	SEC	3	1		4
Operating Systems	MCC	3	1		4
Computer Vision	MSC/UEC	1		2	4

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Pre-requisite Flow Chart

FC

CS101

CS102

CS201

CS202

CS301

241AI007

IC

—

AC

—

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CS102 - Data Structures

CS201 - Object Oriented Programming

CS202 - Database Management Systems

CS301 - Operating Systems

241AI007 - Computer Vision

Advanced Level Courses:

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Course Code	Course Name	Level	L	T	P	S	C	CIE	SEE	Total	Pre-requisite
CS101	Applied Chemistry	Foundation	3		1		4	40	60	100	-
CS102	Data Structures	Foundation	3	1			4	40	60	100	CS101
CS201	Object Oriented Programming	Foundation	3		1		4	40	60	100	CS102
CS301	Operating Systems	Foundation	3	1			4	40	60	100	CS201
Total			12	2	2	0	16				

Minor Stream Courses (MSC) (or) University Open Elective Courses (UEC)

Course Code	Course Name	Level	L	T	P	S	C	CIE	SEE	Total	Pre-requisite
241AI007	Computer Vision	Foundation	1		2	1	4	40	60	100	-
Total			1	0	2	1	4				

Skill Enhancement Courses (SEC)

Course Code	Course Name	Level	L	T	P	S	C	CIE	SEE	Total	Pre-requisite
CS202	Database Management Systems	Foundation	3	1			4	40	60	100	CS102
Total			3	1	0	0	4				

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Semester - 1

Course code	Course Title	Category	Course Credits (L T P S)	Total Credits
CS101	Applied Chemistry	MCC	3 0 1 0 4	4

Semester - 2

Course code	Course Title	Category	Course Credits (L T P S)	Total Credits
CS102	Data Structures	MCC	3 1 0 0 4	4

Semester - 3

Course code	Course Title	Category	Course Credits (L T P S)	Total Credits
CS201	Object Oriented Programming	MCC	3 0 1 0 4	4
CS202	Database Management Systems	SEC	3 1 0 0 4	4

Semester - 4

Course code	Course Title	Category	Course Credits (L T P S)	Total Credits
CS301	Operating Systems	MCC	3 1 0 0 4	4
241AI007	Computer Vision	MSC/UEC	1 0 2 1 4	4

Applied Chemistry
(For Electronics & Communication Engineering)

Course Code: CS101

L T P S C
3 0 1 0 4

Course Outcomes:

At the end of the course, student will be able to:

CO1: Explain the fundamentals of polymers and explore engineering applications

CO2: Interpret the principles and applications of Electrochemical cells

CO3: Summarize the preliminaries of Computational Chemistry and its applications

CO4: Illustrate the fundamentals of nano and smart materials and their applications

CO5: Apply the knowledge of water treatment and E-waste management

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	2	2	1		1	1	1		1		1
CO2		2									
CO3			3	1			2	2			3
CO4	1					2			1		
CO5		2				1	1	1			

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2	PSO3
CO1	2		
CO2		2	
CO3			2
CO4	2		
CO5		2	

methods of polymerization (Suspension polymerization and Emulsion Polymerization)
Difference between thermoplastics and thermosetting plastics; Engineering application of plastics - PTFE and Bakelite
Polymer composites – Composites as structural material; Synthesis and applications of Kevlar and Carbon fibers;
Conducting polymers - Polyacetylene- Mechanism of conduction – applications (polymers in sensors, self cleaning windows)
Biodegradable polymers - Introduction, Polyglycolic acid - synthesis, degradation and uses.

Practice

- 1.Preparation of a polymer (Bakelite)
- 2.Determination of molecular weight of polymer using Ostwald's viscometer

UNIT - II

Electrochemistry and Applications

Electrochemical cell, Nernst equation, cell potential calculations and numerical problems,
Primary cells – Zinc-air battery, Secondary cells – Ni-Cd and, Lead-Acid battery, lithium-ion batteries- working of the batteries including cell reactions; Fuel cells, hydrogen-oxygen,
Methanol-Oxygen fuel cell– working of the cells. Polymer Electrolyte Membrane Fuel cells (PEMFC)
Electrochemical sensors – potentiometric sensors with examples, amperometric sensors with example (Glucose Electrochemical sensor)

Practice

1. Conductometric titration of strong acids strong base
2. Potentiometric titration of red-ox
3. pH -metric titration of Acid-Base
4. Determination of Glucose by Electrochemical sensor

UNIT - III

Computational Chemistry

Computational Quantum Chemistry and its applications, Prediction of Molecular Properties using Computational Chemistry

Molecular Modeling and Structure - molecular modeling today: overview of problems, tools, and solution analysis. Techniques for Conformational Sampling- Monte Carlo, global optimization, etc.

Practice:

1. Molecular docking to predict the interaction energy between molecules

down and bottom up approach) - Sol-gel method. Applications of Carbon nano tubes and Fullerenes, Graphene (water treatment and drug delivery)
Characterization of nanomaterials by SEM and TEM (only Principle).

Smart materials and their engineering applications

Shape memory materials- Poly L-Lactic acid,

Thermoresponse materials – Poly acryl amides , Poly vinyl Amides

Practice:

1. Preparation of nano particles by Green synthesis
2. Characterization of nanoparticles (SEM & TEM)

UNIT - V

Water Treatment & E-Waste Management

Water treatment: Introduction, hardness of water, types, determination of hardness by EDTA method, disadvantages of hard water, removal of hardness by Zeolite and ion exchange method, Desalination of water – reverse osmosis and Electrodialysis.

E-Waste: Introduction, sources of e-waste, Composition, Characteristics, and Need of e-waste management. Toxic materials used in manufacturing electronic and electrical products; Recycling: Recycling of Li-Ion batteries. Extraction of copper from E-waste.

Practice:

1. Determination of Hardness of a ground water sample
2. Determination of Chloride content in given water sample
3. Estimation of dissolved oxygen in given water sample

TEXT BOOKS:

- 1 Chemistry, venkat, vamsi, 3rd

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Data Structures
(For Electronics & Communication Engineering)

Course Code: CS102

L	T	P	S	C
3	1	0	0	4

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Object Oriented Programming
(For Electronics & Communication Engineering)

Course Code: CS201

L	T	P	S	C
3	0	1	0	4

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Database Management Systems
(For Electronics & Communication Engineering)

Course Code: CS202

L	T	P	S	C
3	1	0	0	4

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Operating Systems
(For Electronics & Communication Engineering)

Course Code: CS301

L	T	P	S	C
3	1	0	0	4

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Computer Vision
(Common to CSE, AI/ML & CSE AI)

Course Code: 241AI007

L T P S C
1 0 2 1 4

Course Outcomes:

At the end of the course, student will be able to:

- CO1:** Explain the Fundamentals of Computer Vision and Comprehend Image Formation and Photometric Concepts
- CO2:** Apply the principles of feature detection and matching
- CO3:** Evaluate Computational Photography Techniques and Achieve SuperResolution and Blur Removal and Synthesize Textures.
- CO4:** Analyze basic methods of computer vision related to Structure and Motion
- CO5:** Implement the design of a computer vision system for a 3D Reconstruction, Albedos, image based rendering views and depths.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	2	1	2		2		2		2	1	2
CO2	2	1	2	1	2	1	2		2	1	2
CO3	2		2		2	1	2		2	1	2
CO4	2	1	2	1	2		3	1	1		2
CO5	2		2		2		2		2	1	2

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2	PSO3
CO1	2		2
CO2		1	2
CO3	2		2
CO4		2	
CO5	2		2

Image Processing: Point Operators, Linear Filtering, Global Optimization.

Transformation: Orthogonal, Euclidean, Affine, Projective, etc., Fourier Transform, Convolution and Filtering, Image Enhancement, Restoration, Histogram Processing, Geometric Transformations.

Practice :

1. Implement contrast adjustment of an image. Implement Histogram processing and equalization.
2. Use of Fourier transform for filtering the image

UNIT - II

Feature Detection and Matching:

Points and Patches, Edges, Lines, Segmentation: Active Contours, Split and Merge, Mean Shift and Mode Finding, Normalized Cuts, Feature detectors, Feature descriptors, Feature matching, Feature tracking.

Practice :

1. Implementing image processing operations in python/mat lab/openCV: Reading image, writing image, conversion of images, and complement of an image.
2. Write a program to display of color image.

UNIT - III

Structure and Motion:

Triangular, Twoframe Structure from Motion, Factorization, Bundle Adjustment, Constrained Structure and Motion.

Dense Motion Estimation: Translation Alignment, Parametric Motion, Spline Based Motion.

Practice :

1. Write a program to apply morphological algorithms on an image.
2. Write a program to perform segmentation on an image.

UNIT - IV

Image Stitching:

Motion Analysis, Motion Models, Global Alignment, Composing

Computational Photography: Photometric Calibration, High Dynamic Range Imaging, SuperResolution and Blur Removal, image Matting and Compositing, Texture Analysis and Synthesis.

Practice :

1. Utilization of Texture Analysis and Synthesis.
2. Performing/Implementing Blur Removal and Image Matting using python

Volumetric Representation.

Image based Rendering: View Interpolation, Layered Depth Images, Light Fields and Lumigraphs, Environment Mattes.

Recognition: Object detection, Face detection, Pedestrian detection, Face recognition, Eigenfaces, Active appearance and 3D shape models.

Practice :

1. Write a program to detect edges of an image using Operators.
2. Write a program to apply 2D DFT and DCT on an image and display the result.

TEXT BOOKS:

- 1 CV, Vamsi, venkat, 2nd edition